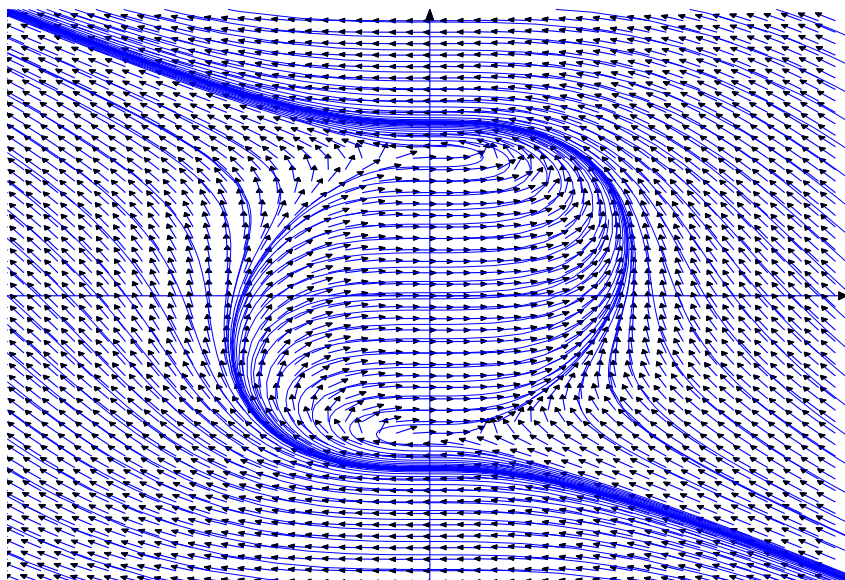


ELEMENTARY DIFFERENTIAL EQUATIONS



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Preface

Elementary Differential Equations with Boundary Value Problems is written for students in science, engineering, and mathematics who have completed calculus through partial differentiation. If your syllabus includes **Chapter 10** (Linear Systems of Differential Equations), your students should have some preparation in linear algebra.

In writing this book I have been guided by the these principles:

- An elementary text should be written so the student can read it with comprehension without too much pain. I have tried to put myself in the student's place, and have chosen to err on the side of too much detail rather than not enough.
- An elementary text can't be better than its exercises. This text includes 1695 numbered exercises, many with several parts. They range in difficulty from routine to very challenging.
- An elementary text should be written in an informal but mathematically accurate way, illustrated by appropriate graphics. I have tried to formulate mathematical concepts succinctly in language that students can understand. I have minimized the number of explicitly stated theorems and definitions, preferring to deal with concepts in a more conversational way, copiously illustrated by 250 completely worked out examples. Where appropriate, concepts and results are depicted in 144 figures.

Although I believe that the computer is an immensely valuable tool for learning, doing, and writing mathematics, the selection and treatment of topics in this text reflects my pedagogical orientation along traditional lines. However, I have incorporated what I believe to be the best use of modern technology, so you can select the level of technology that you want to include in your course. The text includes 336 exercises – identified by the symbols **C** and **C/G** – that call for graphics or computation and graphics. There are also 73 laboratory exercises – identified by **L** – that require extensive use of technology. In addition, several sections include informal advice on the use of technology. If you prefer not to emphasize technology, simply ignore these exercises and the advice.

There are two schools of thought on whether techniques and applications should be treated together or separately. I have chosen to separate them; thus, **Chapter 2** deals with techniques for solving first order equations, and **Chapter 4** deals with applications. Similarly, **Chapter 5** deals with techniques for solving second order equations, and **Chapter 6** deals with applications. However, the exercise sets of the sections dealing with techniques include some applied problems.

Traditionally oriented elementary differential equations texts are occasionally criticized as being collections of unrelated methods for solving miscellaneous problems. To some extent this is true; after all, no single method applies to all situations. Nevertheless, I believe that one idea can go a long way toward unifying some of the techniques for solving diverse problems: variation of parameters. I use variation of parameters at the earliest opportunity in **Section 2.1**, to solve

the nonhomogeneous linear equation, given a nontrivial solution of the complementary equation. You may find this annoying, since most of us learned that one should use integrating factors for this task, while perhaps mentioning the variation of parameters option in an exercise. However, there's little difference between the two approaches, since an integrating factor is nothing more than the reciprocal of a nontrivial solution of the complementary equation. The advantage of using variation of parameters here is that it introduces the concept in its simplest form and focuses the student's attention on the idea of seeking a solution y of a differential equation by writing it as $y = uy_1$, where y_1 is a known solution of related equation and u is a function to be determined. I use this idea in nonstandard ways, as follows:

- In [Section 2.4](#) to solve nonlinear first order equations, such as Bernoulli equations and nonlinear homogeneous equations.
- In [Chapter 3](#) for numerical solution of semilinear first order equations.
- In [Section 5.2](#) to avoid the necessity of introducing complex exponentials in solving a second order constant coefficient homogeneous equation with characteristic polynomials that have complex zeros.
- In [Sections 5.4, 5.5, and 9.3](#) for the method of undetermined coefficients. (If the method of annihilators is your preferred approach to this problem, compare the labor involved in solving, for example, $y'' + y' + y = x^4 e^x$ by the method of annihilators and the method used in [Section 5.4](#).)

Introducing variation of parameters as early as possible ([Section 2.1](#)) prepares the student for the concept when it appears again in more complex forms in [Section 5.6](#), where reduction of order is used not merely to find a second solution of the complementary equation, but also to find the general solution of the nonhomogeneous equation, and in [Sections 5.7, 9.4, and 10.7](#), that treat the usual variation of parameters problem for second and higher order linear equations and for linear systems.

You may also find the following to be of interest:

- [Section 2.6](#) deals with integrating factors of the form $\mu = p(x)q(y)$, in addition to those of the form $\mu = p(x)$ and $\mu = q(y)$ discussed in most texts.
- [Section 4.4](#) makes phase plane analysis of nonlinear second order autonomous equations accessible to students who have not taken linear algebra, since eigenvalues and eigenvectors do not enter into the treatment. Phase plane analysis of constant coefficient linear systems is included in [Sections 10.4–6](#).
- [Section 4.5](#) presents an extensive discussion of applications of differential equations to curves.
- [Section 6.4](#) studies motion under a central force, which may be useful to students interested in the mathematics of satellite orbits.
- [Sections 7.5–7](#) present the method of Frobenius in more detail than in most texts. The approach is to systematize the computations in a way that avoids the necessity of substituting the unknown Frobenius series into each equation. This leads to efficiency in the computation of the coefficients of the Frobenius solution. It also clarifies the case where the roots of the indicial equation differ by an integer ([Section 7.7](#)).

- The free Student Solutions Manual contains solutions of most of the even-numbered exercises.
- The free Instructor's Solutions Manual is available by email to wtrench@trinity.edu, subject to verification of the requestor's faculty status.

The following observations may be helpful as you choose your syllabus:

- Section 2.3 is the only specific prerequisite for Chapter 3. To accomodate institutions that offer a separate course in numerical analysis, Chapter 3 is not a prerequisite for any other section in the text.
- The sections in Chapter 4 are independent of each other, and are not prerequisites for any of the later chapters. This is also true of the sections in Chapter 6, except that Section 6.1 is a prerequisite for Section 6.2.
- Chapters 7, 8, and 9 can be covered in any order after the topics selected from Chapter 5. For example, you can proceed directly from Chapter 5 to Chapter 9.
- The second order Euler equation is discussed in Section 7.4, where it sets the stage for the method of Frobenius. As noted at the beginning of Section 7.4, if you want to include Euler equations in your syllabus while omitting the method of Frobenius, you can skip the introductory paragraphs in Section 7.4 and begin with Definition 7.4.2. You can then cover Section 7.4 immediately after Section 5.2.

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